

Operation Staff-WiFi

Two Students, One Password, Zero Bars

The Story

It is the night before the end-semester exam. You are in your hostel room at **IIIT Vadodara**, trying to watch a last-minute lecture on YouTube. But the student WiFi is doing what it does best—buffering. The video loads one frame per minute. Your notes are on Google Drive. Google Drive says “Checking connection...” It has been checking for 40 minutes.

Meanwhile, you notice that the **Staff WiFi** next door has **full bars**. Five glorious bars. Fast, reliable, and completely **password-protected**.

You and your friend have a plan. To make the task easier, you **split in two**. Neither of you can solve it alone. You must **exchange values**—and each computation involves billions of iterations.

The Task

Write a **distributed program** that simulates two students working together.

- Extract A and B from student IDs (last 4 digits each).
- Implement the **grinder function**—each searches for values up to $N = 2,000,000,000$.
- Exchange intermediate values between workers.
- Both workers independently compute the verification value—they **must match**.
- One worker produces the final 4-digit password.
- Measure total computation time.

Use appropriate functions for point-to-point communication. Include both student IDs in comments.

The Cipher

Setup: Two student IDs $\rightarrow$ last 4 digits $\rightarrow$ keys A and B .

The Grinder: $\mathcal{G}(\text{seed}, m, a, \text{mod})$ performs:

$$r_0 = \text{seed} \quad r_{i+1} = (r_i \times m + a) \bmod \quad \mathcal{G} = r_N$$

where $N = 2,000,000,000$. Each call is computationally expensive.

The Cipher Sequence (all mod 9973 unless noted):

Step	Formula	Who
1a	$\alpha' = \mathcal{G}(A, 31, 17, 9973)$	Student 1
1b	$\alpha'' = \mathcal{G}(\alpha', 37, 11, 9973)$	Student 1
1c	$\alpha = (\alpha' + \alpha'') \bmod 9973$	Student 1
α is transmitted $\rightarrow$		
2a	$\beta' = \mathcal{G}(\alpha, B, 13, 9973)$	Student 2
2b	$\beta'' = \mathcal{G}(\beta', 41, 19, 9973)$	Student 2
2c	$\beta = (\beta' + \beta'') \bmod 9973$	Student 2
β is transmitted $\leftarrow$		
3	Verify = $\mathcal{G}(\alpha + \beta, 7, 3, 101)$	Both
4	$R = \mathcal{G}(\alpha \times \beta + A + B, 13, 7, 9973)$	Student 1
5	Password = $R \bmod 10000$	Student 1